Walmart Business Problems Solutions

1. Analyze Payment Methods and Sales

- **Question:** What are the different payment methods, and how many transactions and items were sold with each method?

payment_method	number_of_transactions	quantity_sold
Ewallet	3881	8932
Cash	1832	4984
Credit card	4256	9567

- **Purpose:** This helps understand customer preferences for payment methods, aiding in payment optimization strategies.

2. Identify the Highest-Rated Category in Each Branch

- **Question:** Which category received the highest average rating in each branch?

branch	category	average_rating	rank_num
WALM001	Electronic accessories	7.45	1
WALM002	Food and beverages	8.25	1
WALM003	Sports and travel	7.5	1
WALM004	Food and beverages	9.3	1
WALM005	Health and beauty	8.366666666666667	1
WALM006	Fashion accessories	6.797058823529412	1
WALM007	Food and beverages	7.55	1
WALM008	Food and beverages	7.4	1
WALM009	Sports and travel	9.6	1
WALM010	Electronic accessories	9	1
WALM011	Food and beverages	7	1
WALM012	Health and beauty	7.45	1

- **Purpose:** This allows Walmart to recognize and promote popular categories in specific branches, enhancing customer satisfaction and branch-specific marketing.

3. Determine the Busiest Day for Each Branch

- **Question:** What is the busiest day of the week for each branch based on transaction volume?

branch	day_name	no_transactions	rank_num
WALM001	Thursday	16	1
WALM002	Thursday	15	1
WALM003	Tuesday	33	1
WALM004	Sunday	14	1
WALM005	Wednesday	19	1
WALM006	Thursday	15	1
WALM007	Friday	12	1
WALM007	Sunday	12	1
WALM008	Tuesday	17	1
WALM009	Sunday	42	1
WALM010	Wednesday	12	1
WALM011	Tuesday	18	1

- **Purpose:** This insight helps in optimizing staffing and inventory management to accommodate peak days.

4. Analyze Category Ratings by City

- **Question:** What are the average, minimum, and maximum ratings for each category in each city?

city	category	max_rating	min_rating	avg_rating
San Antonio	Health and beauty	9.1	5	7.05
Harlingen	Electronic accessories	9.6	9.6	9.6
Haltom City	Home and lifestyle	9.5	3	6.23
Bedford	Health and beauty	9.3	6.1	8.15
Irving	Sports and travel	5.3	5.3	5.3
Denton	Electronic accessories	9	4.1	6.7
Cleburne	Electronic accessories	7.8	5.8	7.25
Canyon	Home and lifestyle	9	3	6.25
Grapevine	Health and beauty	7.2	7.2	7.2
Texas City	Food and beverages	5.9	5.5	5.7
Irving	Fashion accessories	9.8	3	6.21
San Angelo	Electronic accessories	7	3	5.83

- **Purpose:** This data can guide city-level promotions, allowing Walmart to address regional preferences and improve customer experiences.

5. Calculate Total Profit by Category

- **Question:** What is the total profit for each category, ranked from highest to lowest?

category	total_revenue	total_profit
Fashion accessories	489480.89999999997	192314.89
Home and lifestyle	489250.06	192213.64
Electronic accessories	78175.02999999998	30772.49
Food and beverages	53471.28000000006	21552.86
Sports and travel	52497.93000000002	20613.81
Health and beauty	46851.17999999998	18671.73

- **Purpose:** Identifying high-profit categories helps focus efforts on expanding these products or managing pricing strategies effectively.

6. Determine the Most Common Payment Method per Branch

- **Question:** What is the most frequently used payment method in each branch?

branch	payment_method	total_payment_method	rank_num
WALM001	Ewallet	45	1
WALM002	Ewallet	37	1
WALM003	Credit card	115	1
WALM004	Ewallet	44	1
WALM005	Ewallet	56	1
WALM006	Ewallet	50	1
WALM007	Ewallet	52	1
WALM008	Ewallet	39	1
WALM009	Credit card	139	1
WALM010	Ewallet	47	1
WALM011	Ewallet	39	1
WALM012	Ewallet	52	1

- **Purpose:** This information aids in understanding branch-specific payment preferences, potentially allowing branches to streamline their payment processing systems.

7. Analyze Sales Shifts Throughout the Day

- **Question:** How many transactions occur in each shift (Morning, Afternoon, Evening) across branches?

branch	day_time	count(*)
WALM001	Afternoon	36
WALM001	Evening	30
WALM001	Morning	8
WALM002	Afternoon	29
WALM002	Evening	21
WALM002	Morning	15
WALM003	Afternoon	95
WALM003	Morning	50
WALM003	Evening	41
WALM004	Afternoon	27
WALM004	Evening	24
WALM004	Morning	9

- **Purpose:** This insight helps in managing staff shifts and stock replenishment schedules, especially during high-sales periods.

8. Identify Branches with Highest Revenue Decline Year-Over-Year

- **Question:** Which branches experienced the largest decrease in revenue compared to the previous year?

branch	last_year_revenue	current_year_revenue	decrease_ratio
WALM045	1731	647	62.62
WALM047	2581	1069	58.58
WALM098	2446	1030	57.89
WALM033	2099	931	55.65
WALM081	1723	850	50.67

- **Purpose:** Detecting branches with declining revenue is crucial for understanding possible local issues and creating strategies to boost sales or mitigate losses.